

Angle Properties – Answer Sheet

WorksheetHub AI
Mathematics · Grade 7

Unit 3 · Geometry

Medium

Angles in lines, triangles and parallel lines

A. Find the missing angle x

- 1 $x = 180 - 65 = 115^\circ$
- 2 $x = 180 - 52 - 67 = 61^\circ$
- 3 $x = 360 - 120 - 95 = 145^\circ$
- 4 Vertically opposite angles are equal: 72°

B. Parallel lines (use the parallel-lines picture)

- 5 $a = 65^\circ$, corresponding angles
- 6 $b = 180 - 65 = 115^\circ$ (angles on a straight line with a)
- 7 $c = 115^\circ$, alternate angles with b

C. Polygons

- 8 $(5 - 2) \times 180 = 540^\circ$
- 9 $(6 - 2) \times 180 = 720^\circ$
- 10 $720 \div 6 = 120^\circ$

★ Challenge

- 11 $6x = 180$, $x = 30^\circ$; angles are 30° , 60° , 90°
- 12 Exterior angles add to 360° : $360 \div 45 = 8$ sides